

Preston Beachum

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Education

University of North Carolina at Wilmington Expected May 2027
Bachelor of Science in Software Engineering GPA: 3.5
Relevant Coursework: Software Construction, Design and Analysis of Algorithms, Object-Oriented Programming (Java), Web Development (HTML/CSS, JS), Software Systems Analysis, Architecture & Design Patterns

Work Experience

Associate Software Engineer — Ohanafy Jun 2026 – Present — Wilmington, NC

- Designed and shipped a customer-facing branding customization feature, enabling clients to apply their own brand identity directly within their production Salesforce orgs

Software Engineering Intern — Ohanafy Feb 2026 – May 2026 — Wilmington, NC

- Engineered and shipped production-grade features by designing and implementing custom data models (objects, fields) aligned with evolving business requirements
- Built and integrated Lightning Web Components (LWC) with Apex backend services, enabling full-stack functionality across UI and business logic layers
- Executed end-to-end regression testing across core workflows, identifying and documenting critical defects pre-release to ensure system stability
- Refactored frontend components by migrating legacy LWCs to Tailwind CSS, improving UI consistency and development velocity
- Contributed to system design discussions, influencing component architecture, scalability, and long-term maintainability

Systems Engineer Intern — Union County Public Schools (UCPS) May 2025 – Aug 2025 — Monroe, NC

- Automated 10+ internal workflows and system processes, improving operational efficiency by approximately 60% and reducing manual overhead
- Developed and maintained internal web-based tools supporting systems integration and end-user infrastructure
- Standardized repeatable tasks through scripting and documentation, improving onboarding speed and system reliability

Research Experience

Undergraduate Research Assistant — INSIGHTs Network Visualization Project Aug 2025 – Present
UNCW School of Nursing & Department of Computer Science

- Selected as the only Software Engineering student featured in UNCW Research Magazine for technical contributions to a research-grade medical visualization platform
- Designed and prototyped next-generation system architecture using real-world usage data and clinician feedback
- Led architectural refactoring across 3 Flask-based microservices (INSIGHTs-GBA, PROACTIVE, SIMU), improving scalability, modularity, and data flow
- Collaborated with cross-functional clinical and academic teams to deliver systems evaluated in real-world healthcare environments

Undergraduate Research Assistant — LLM Security Research Lab May 2026 – Present
UNCW Department of Computer Science

- Researched LLM security and vulnerability detection workflows using AWS Bedrock and related AWS services to evaluate and compare transformer-based AI models.
- Built early-stage evaluation pipelines for testing model outputs, response consistency, and comparative behavior across multiple foundation models.
- Conducted experimentation with prompt engineering, adversarial prompting, and model robustness testing in cloud-based environments.
- Collaborated on technical analysis of model security tradeoffs, inference behavior, and practical considerations for secure AI system deployment.

Honors & Recognition

- Selected to present engineering research and program outcomes to the UNCW Board of Trustees, representing the College of Science & Engineering alongside faculty leadership
- Selected to be featured in a national documentary by the American Society for Engineering Education (ASEE), representing UNCW Engineering through on-camera interview and program showcase

Projects

Wi-Fi Analytics Platform — Network telemetry collection and web dashboard

- Architected and implemented a full-stack network telemetry platform processing thousands of 802.11 packets per minute from distributed edge devices
- Built a Raspberry Pi-based capture system operating in monitor mode to collect probe requests, beacon frames, and traffic metadata
- Designed backend services to stream, store, and analyze structured packet data using PostgreSQL and Node.js
- Automated device provisioning and runtime behavior using Bash and systemd, enabling zero-touch deployment into packet capture mode
- Developed a web-based dashboard (React/Next.js) for real-time visualization and analysis of network activity

Collaborative Task Manager — Scalable real-time task management platform

- Engineered a scalable backend using Spring Boot and PostgreSQL, supporting secure user authentication and role-based access control
- Implemented a scheduled notification system achieving 100% delivery success during integration testing
- Automated API workflows using Bash scripts, reducing manual testing and provisioning time by 80%
- Designed RESTful APIs supporting concurrent users and real-time task updates

Technical Skills

Languages: Java, JavaScript, Python, HTML/CSS

Frameworks & Libraries: React, Next.js, Spring Boot, Flask, Node.js, Next.js

Databases: PostgreSQL, SQLite, SQL

Tools & Technologies: Git, Docker, Linux, Bash, WebSocket, npm, AWS, ESLint, GitHub Actions